Kepler's Laws from the Newton–Euler Laws and Homothetic Symmetry

Equal areas, conic orbits, and $T^2 \propto a^3$ from a scaling of trajectories

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Statement and assumptions

Kepler's laws

1. **Orbits.** A bound planet moves on an ellipse with the Sun at one focus.
2. **Areas.** The radius vector sweeps equal areas in equal times.
3. **Periods.** T^2/a^3 is the same for every orbit about the same Sun (a = semi-major axis).

Assumptions

- **Euler's first law** (Newton's second): $m\ddot{\boldsymbol{r}} = \boldsymbol{F}$.
- **Euler's second law**: $\dfrac{d}{dt}(\boldsymbol{r} \times m\dot{\boldsymbol{r}}) = \boldsymbol{r} \times \boldsymbol{F}$.
- **Gravitation**: $\boldsymbol{F} = -\dfrac{GMm}{\|\boldsymbol{r}\|^3}\,\boldsymbol{r}$ (central, inverse square).

Two bodies reduce to one: with $\boldsymbol{r}$ the relative position and reduced mass μ, the equation of motion is

$$\ddot{\boldsymbol{r}} = -\frac{k}{\|\boldsymbol{r}\|^3}\,\boldsymbol{r}, \qquad k = G(M+m). \tag{$*$}$$

Throughout, $r = \|\boldsymbol{r}\|$ and $\boldsymbol{\ell} = \boldsymbol{r} \times \dot{\boldsymbol{r}}$ is the specific angular momentum.

Second law: equal areas

Proposition

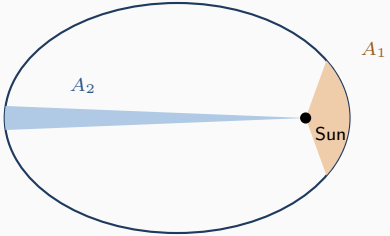
$\boldsymbol{\ell} = \boldsymbol{r} \times \dot{\boldsymbol{r}}$ is constant. Hence the motion lies in the plane through the origin orthogonal to $\boldsymbol{\ell}$, and the areal velocity is $\dfrac{dA}{dt} = \tfrac{1}{2}\|\boldsymbol{\ell}\|$.

Proof. By Euler's second law with a central force,

$$\dot{\boldsymbol{\ell}} = \dot{\boldsymbol{r}} \times \dot{\boldsymbol{r}} + \boldsymbol{r} \times \ddot{\boldsymbol{r}} = \boldsymbol{r} \times \left(-\tfrac{k}{r^3}\boldsymbol{r} \right) = \boldsymbol{0}.$$

Since $\boldsymbol{r} \cdot \boldsymbol{\ell} = 0$ for all t, $\boldsymbol{r}(t)$ stays in the plane $\boldsymbol{\ell}^{\perp}$. In time dt the radius vector sweeps the triangle spanned by $\boldsymbol{r}$ and $\dot{\boldsymbol{r}}\,dt$, of area $dA = \tfrac{1}{2}\|\boldsymbol{r} \times \dot{\boldsymbol{r}}\|\,dt = \tfrac{1}{2}\|\boldsymbol{\ell}\|\,dt$. $\square$

Only centrality is used here, not the inverse-square law.



equal $\Delta t \;\Rightarrow\; A_1 = A_2$

First law: conic orbits via the Laplace–Runge–Lenz vector

Proposition

$\boldsymbol{A} = \dot{\boldsymbol{r}} \times \boldsymbol{\ell} - k\,\dfrac{\boldsymbol{r}}{r}$ is constant, and the orbit is the conic

$$r(\theta) = \frac{p}{1 + e\cos\theta}, \qquad p = \frac{\|\boldsymbol{\ell}\|^2}{k}, \quad e = \frac{\|\boldsymbol{A}\|}{k},$$

with θ measured from $\boldsymbol{A}$. For bound motion ($e < 1$) this is an ellipse with the origin at a focus.

Proof. Using $(*)$, $\dot{\boldsymbol{\ell}} = 0$ and $\boldsymbol{a} \times (\boldsymbol{b} \times \boldsymbol{c}) = \boldsymbol{b}(\boldsymbol{a}\cdot\boldsymbol{c}) - \boldsymbol{c}(\boldsymbol{a}\cdot\boldsymbol{b})$:

$$\ddot{\boldsymbol{r}} \times \boldsymbol{\ell} = -\frac{k}{r^3}\,\boldsymbol{r} \times (\boldsymbol{r} \times \dot{\boldsymbol{r}}) = \frac{k}{r^3}\big(r^2\dot{\boldsymbol{r}} - (\boldsymbol{r}\cdot\dot{\boldsymbol{r}})\boldsymbol{r}\big) = k\,\frac{d}{dt}\Big(\frac{\boldsymbol{r}}{r}\Big),$$

so $\dot{\boldsymbol{A}} = 0$. Dotting $\boldsymbol{A}$ with $\boldsymbol{r}$ and using $\boldsymbol{r}\cdot(\dot{\boldsymbol{r}} \times \boldsymbol{\ell}) = \boldsymbol{\ell}\cdot(\boldsymbol{r} \times \dot{\boldsymbol{r}}) = \|\boldsymbol{\ell}\|^2$:

$$\|\boldsymbol{A}\|\, r\cos\theta = \boldsymbol{A}\cdot\boldsymbol{r} = \|\boldsymbol{\ell}\|^2 - kr \quad\Longrightarrow\quad r\,(k + \|\boldsymbol{A}\|\cos\theta) = \|\boldsymbol{\ell}\|^2. \qquad \square$$

For the ellipse: $a = \dfrac{p}{1-e^2}$, $\quad b = a\sqrt{1-e^2}$, $\quad$ hence $p = \dfrac{b^2}{a}$.

The homothetic symmetry of $(*)$

Lemma (homothety of trajectories)

If $\boldsymbol{r}(t)$ solves $(*)$, then for every $\lambda > 0$

$$\boldsymbol{r}_\lambda(t) = \lambda\, \boldsymbol{r}\big(\lambda^{-3/2} t\big)$$

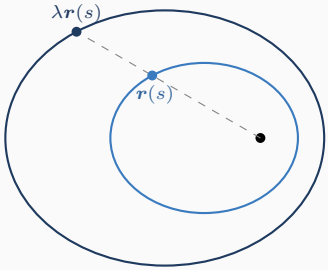
also solves $(*)$.

Proof. With $s = \lambda^{-3/2} t$, the chain rule gives

$$\ddot{\boldsymbol{r}}_\lambda(t) = \lambda \cdot \lambda^{-3}\, \ddot{\boldsymbol{r}}(s) = -\lambda^{-2}\, \frac{k\, \boldsymbol{r}(s)}{\|\boldsymbol{r}(s)\|^3}, \quad \text{and}$$

$$-\frac{k\, \boldsymbol{r}_\lambda(t)}{\|\boldsymbol{r}_\lambda(t)\|^3} = -\frac{k\, \lambda\, \boldsymbol{r}(s)}{\lambda^3 \|\boldsymbol{r}(s)\|^3} = -\lambda^{-2}\, \frac{k\, \boldsymbol{r}(s)}{\|\boldsymbol{r}(s)\|^3}. \qquad \square$$

Lengths scale by λ and times by $\lambda^{3/2}$; the orbit's shape (eccentricity) is unchanged.



same focus, same e;
$a \mapsto \lambda a, \quad T \mapsto \lambda^{3/2} T$

Third law from homothety

Step 1: scaling alone

If $\boldsymbol{r}$ is a bound orbit with semi-major axis a, eccentricity e, period T, then $\boldsymbol{r}_\lambda$ has semi-major axis λa, the same e, and period $\lambda^{3/2}T$. Hence $\dfrac{(\lambda^{3/2}T)^2}{(\lambda a)^3} = \dfrac{T^2}{a^3}$: the ratio depends only on the shape, $T^2/a^3 = C(e)$.

Step 2: C does not depend on e

Second law: one revolution sweeps area πab at rate $\frac{1}{2}\|\boldsymbol{\ell}\|$, so $T = 2\pi ab/\|\boldsymbol{\ell}\|$. First law: $\|\boldsymbol{\ell}\|^2 = kp = kb^2/a$. Hence

$$T = \frac{2\pi ab}{b\sqrt{k/a}} = \frac{2\pi}{\sqrt{k}}\, a^{3/2}, \qquad \boxed{\frac{T^2}{a^3} = \frac{4\pi^2}{G(M+m)}}$$

Symmetry fixes the exponent $3/2$; the first two laws fix the constant (the same for all planets when $m \ll M$).

The general principle: mechanical similarity

Homogeneous potentials [Landau & Lifshitz, Mechanics, §10]

If $U(\lambda \boldsymbol{r}) = \lambda^\nu U(\boldsymbol{r})$, then $\boldsymbol{r}_\lambda(t) = \lambda\, \boldsymbol{r}(\lambda^{-(1-\nu/2)} t)$ maps solutions of $m\ddot{\boldsymbol{r}} = -\nabla U$ to solutions. Characteristic times scale as

$$\frac{t'}{t} = \left(\frac{\ell'}{\ell}\right)^{1-\nu/2}.$$

system	ν	$t \propto \ell^{1-\nu/2}$	consequence
harmonic oscillator, $U \propto r^2$	2	ℓ^0	period independent of amplitude
uniform field, $U \propto z$	1	$\ell^{1/2}$	fall time $\propto \sqrt{h}$
Kepler, $U \propto -1/r$	-1	$\ell^{3/2}$	$T^2 \propto a^3$

Summary

- **Second law** follows from Euler's second law and the centrality of gravity: $\boldsymbol{\ell}$ is conserved, and $dA/dt = \frac{1}{2}\|\boldsymbol{\ell}\|$.
- **First law** follows from conservation of the Laplace–Runge–Lenz vector, which is specific to the inverse-square force.
- **Third law**: the equation of motion is invariant under $\boldsymbol{r} \mapsto \lambda \boldsymbol{r}$, $t \mapsto \lambda^{3/2} t$. This gives $T^2/a^3 = C(e)$ immediately; the first two laws show $C = 4\pi^2/G(M+m)$.
- The scaling argument is an instance of mechanical similarity for homogeneous potentials.